12.DENGUE FEVER

Overview

Dengue fever is an acute viral illness associated with fever and a range of symptoms including nausea, vomiting, rash, muscle pain, headache, retro-orbital pain, joint pain and, in some cases, hemorrhagic manifestations.

The ICMR STW is designed to help clinicians identify **probable dengue**, recognize **warning signs**, identify **severe dengue**, investigate appropriately and manage complications such as shock.

When to Suspect Dengue

Dengue should be suspected when a child:

- Has **fever**
- Has **at least two** of the following:
    - Nausea
    - Vomiting
    - Rash
    - Myalgia
    - Headache
    - Retro-orbital pain
    - Arthralgia
    - Hemorrhagic manifestations
- Lives in or has travelled to a **dengue-endemic area**.

Simple Recognition Rule

Fever + ≥2 compatible symptoms + exposure to a dengue-endemic area → Suspect dengue

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Symptoms

Common symptoms/signs include:

- **Fever**

- Nausea
- Vomiting
- Rash
- Muscle pain (**myalgia**)
- Headache
- **Pain behind the eyes (retro-orbital pain)**
- Joint pain (**arthralgia**)
- Hemorrhagic manifestations

These symptoms are the clinical criteria highlighted in the ICMR STW for suspecting dengue.

Warning Signs

A patient with dengue should be carefully assessed for **warning signs**, including:

- **Abdominal pain or tenderness**
- **Persistent vomiting**
- Clinical fluid accumulation:
    - Pleural effusion
    - Ascites
- **Mucosal bleeding**, such as:
    - Melena
    - Epistaxis
    - Gum bleeding
- **Liver enlargement >2 cm**
- Signs of shock, including:
    - Weak, rapid pulse
    - Pulse pressure <20 mmHg
    - Hypotension
    - Cold, clammy skin

    - Restlessness
- Rising PCV with thrombocytopenia, including situations where clinical symptoms may not yet be prominent.

Important

Dengue + warning signs → requires closer monitoring and appropriate medical management.

Severe Dengue

The ICMR STW identifies severe dengue through serious complications such as:

1. Severe fluid accumulation

Particularly when associated with:

- **Respiratory distress**

2. Severe bleeding

Significant hemorrhagic manifestations may require urgent intervention.

3. Impaired consciousness

Altered or impaired consciousness is a serious sign requiring urgent medical attention.

Investigations

The ICMR STW lists investigations according to clinical requirements.

Essential investigations

- **Hemoglobin (Hb)**
- **Total leukocyte count (TLC)**
- **Differential leukocyte count (DLC)**
- **Platelet count**
- **Packed cell volume (PCV)**
- Tourniquet test
- **NS1 antigen — ELISA**
- Chest X-ray
- Liver function tests (LFT)

- Renal function tests (RFT)
- Creatine phosphokinase (CPK)
- Albumin
- Ultrasound abdomen
- **Dengue IgM**

Additional investigations when required

- Dengue PCR
- Echocardiography
- Central venous pressure (CVP) monitoring
- Ultrasound-guided assessment of IVC collapsibility for monitoring hypovolemia.

Important: Do not present all of these as tests that every dengue patient automatically needs. The STW categorizes investigations as **essential, desirable and optional**, depending on the clinical situation.

Treatment

Probable Dengue WITHOUT Warning Signs

The ICMR STW recommends:

- **Symptomatic ambulatory treatment**
- **Paracetamol for fever**
- **Avoid NSAIDs**
- Daily monitoring of:
    - Clinical condition
    - PCV
    - Platelet count.

Dengue WITH Warning Signs

Patients with warning signs require closer clinical management and monitoring.

Particular attention should be given to:

- Fluid status

- Hemodynamic status
- PCV
- Platelet count
- Urine output
- Development of shock
- Respiratory distress

If the patient's condition deteriorates, management should progress according to the shock/severe-dengue pathway.

Dengue With Shock

The ICMR workflow distinguishes between:

Compensated Shock

Features may include:

- Tachypnea
- Tachycardia
- Blood pressure may still be maintained/normotensive.

Hypotensive Shock

Features include:

- Tachypnea
- Tachycardia
- Hypotension
- Peripheral pulses may not be palpable.

Emergency Management

The STW directs clinicians to:

Assess airway, breathing and circulation → Start oxygen → Initiate appropriate fluid resuscitation → Continuously monitor → Urgent referral to a facility with PICU capability when required.

Fluid Management in Shock

The ICMR STW provides a **stepwise fluid-management pathway** based on clinical response and monitoring.

The document specifies:

- Initial crystalloid/colloid bolus according to the shock pathway.
- Reassessment using parameters such as:
    - Heart rate
    - Respiratory rate
    - Blood pressure
    - PCV
    - CVP where applicable
- Fluid infusion is progressively reduced when the patient improves.
- If there is no improvement, clinicians should assess for:
    - Blood loss
    - Acidosis
    - Cardiac dysfunction
    - Fluid overload
    - Other causes of persistent shock.

For your awareness PDF, do NOT reproduce all the exact fluid rates unless your template has enough space. The complete fluid algorithm is already available in the official ICMR STW and is more appropriate for healthcare professionals than for a general awareness chatbot.

Blood Products

The ICMR workflow provides indications for blood-product use.

Packed Red Cells

The STW lists situations including:

- Significant overt blood loss
- Refractory shock
- Other clinically significant blood-loss situations.

Platelets

The STW lists situations including:

- Prolonged shock
- Prophylactic platelet transfusion at very low platelet counts
- Systemic massive bleeding.

Fresh Frozen Plasma / Cryoprecipitate

May be considered when there is:

- **Coagulopathy with bleeding**.

Monitoring

During significant dengue illness, the ICMR workflow emphasizes monitoring of:

- Heart rate
- Respiratory rate
- Blood pressure
- PCV
- Platelet count
- Fluid status
- Clinical condition
- Urine output
- Respiratory status

In shock management, reassessment is performed repeatedly to determine whether the patient is improving or requires escalation of treatment.

Discharge Criteria

The ICMR STW specifies that **all** of the following should be present:

Clinical

- **No fever for 48 hours**
- Improvement in overall clinical condition

- Improvement in general well-being
- Improving appetite
- Stable hemodynamic status
- Adequate urine output
- No significant respiratory distress

Laboratory

- **Increasing trend in platelet count**
- **Stable haematocrit without intravenous fluids**.

The workflow indicates that once the patient is stable, observation should continue and discharge can be considered when the discharge criteria are fulfilled.

Quick Fact Sheet

Feature	**Dengue Fever**
ICMR source	Standard Treatment Workflow
ICD-10	A90
When to suspect	Fever + ≥2 compatible symptoms + exposure to dengue-endemic area
Key symptoms	Fever, nausea, vomiting, rash, myalgia, headache, retro-orbital pain, arthralgia
Important warning signs	Abdominal pain, persistent vomiting, fluid accumulation, mucosal bleeding, liver enlargement, shock
Severe dengue	Severe fluid accumulation with respiratory distress, severe bleeding or impaired consciousness
Important investigations	CBC/PCV, platelets, NS1, IgM and other tests as clinically indicated
Treatment without warning signs	Symptomatic ambulatory treatment + paracetamol; avoid NSAIDs
Severe disease/shock	Urgent stabilization, fluid management, monitoring and referral/PICU when required

Feature	Dengue Fever
Discharge	Afebrile 48 h + clinical improvement + increasing platelets + stable haematocrit without IV fluids

Official reference

Indian Council of Medical Research (ICMR), Department of Health Research, Ministry of Health & Family Welfare, Government of India. *Standard Treatment Workflow (STW) for the Management of Dengue Fever, ICD-10-A90*, October 2019.

The official ICMR page lists **Dengue fever** under its Paediatrics Standard Treatment Workflow.